

Manuscript_Analyses

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Setup

Data Retrieval

All data is hosted on github. Please see the following for information on how to retrieve each data set used in the analysis.

App Comparison Data Excel sheet with e a table comparing the different R shiny applications availabel for batche effect analysis.

```
app_comparison_location <- "https://raw.githubusercontent.com/jessmcc22/lab_meeting/refs/heads/devel/Ma
app_comparison_data <- read.csv(app_comparison_location, header = TRUE)
```

ADNI Data Alzheimer's imaging data provided by Taki Shinohara (UPenn). In this data set, three different scanners were used (Siemens Healthineers (n=96), Philips Medical Systems (n = 96) or GE Healthcare (n = 287)) and in a previous analysis done with DeepComBat, a batch effect was seen between Siemens and Philips/GE (combined as one batch) scanners. We are hoping to recreate this batch effect with BatchQC when comparing Alzheimer diagnosis (cognitively normal (CN), late mild cognitive impairment (LMCI) and Alzheimer disease (AD)). To note, this data includes up to 6 images for each subject taken at different visits, thus there are 2515 unique images.

```
ADNI_data_location <- "https://raw.githubusercontent.com/ntustison/CrossLong/refs/heads/master/Data/rec
ADNI_additional_metadata_location <- "https://raw.githubusercontent.com/jessmcc22/lab_meeting/refs/head
ADNI_data <- read.csv(ADNI_data_location, header = TRUE)
ADNI_additional_metadata <- read.csv(ADNI_additional_metadata_location,
  header = TRUE)
```

Introduction

Figure 1: R Shiny Batch Effect Application Comparison

Many different R Shiny apps exist that streamline data exploration, analysis and visualization of batch correction analyses. The methods and tools available in some common applications are compared in the below table. The information in this table was obtained from the accompanying manuscript publication for each application. The table presented in our paper was formatted in excel and a simpler, unformatted table with the same information is presented below.

```
knitr::kable(app_comparison_data)
```

X	BatchQC	Batch.FLEX	Batch.Server	DE.Browser
Batch Corr. Methods				
SVA	x	x	x	
ComBat	x	x		x
autoComBat			x	
ComBat-Seq	x	x		
limma	x	x		
Harman		x		x
RUVg		x		
correctBatch		x		
exploBatch				
Visulaizations				
PCVA		x	x	
UMAP	x	x	x	
PCA	x	x		x
Heatmaps	x	x		x
Dendogram	x			
Circular Dendogram	x			
Volcano Plot	x			x
Scree Plot		x		
All2all scatter plots				x
IQR and Density Plots	x			x
Evalutaions				
Variation Analysis	x		x	
Differential Expression	x			x
Distribution Check	x			
Distribution Check	x			
Additional Features				
Shiny Interface	x	x	x	x
Standalone R Functions	x	x		
Download Data	x	x	x	
Normalization	x	x		x

Methods

Lambda Statistic

Model Selection Checks

Data Examples

ADNI Data The original data set includes multiple images for each subject taken in a longitudinal manner at multiple different visits. For our analysis, we are interested in comparing the baseline images at the first visit. The set of images also contain images of various locations in the brain and for this analysis, we will focus on the temporal location.

```
VISIT_NUMBER <- 0
BRAIN_LOCATION <- "temporal"
```

Tidy the downloaded file into metadata and cortical thickness dataframes

```
ADNI_data_t0 <- ADNI_data[order(ADNI_data$IMAGE_ID),] %>%
  filter(VISIT == VISIT_NUMBER)
Image_ID <- as.character(ADNI_data_t0$IMAGE_ID)
ADNI_metadata <- as.data.frame(ADNI_data_t0[,1:8], row.names = Image_ID)
cortical_thickness <- (as.data.frame(ADNI_data_t0[,9:70], row.names = Image_ID))
```

Subset cortical thickness measures based on location of interest.

```
specific_location_data <- cortical_thickness %>%
  select(contains(BRAIN_LOCATION)) %>%
  t()
cortical_thickness <- t(cortical_thickness)
```

Tidy the data by creating a column name that contains all the scan locations and a column name with the thickness values

```
tidy_ADNI <- ADNI_data_t0 %>%
  pivot_longer(9:dim(ADNI_data_t0)[2], names_to = "LOCATION", values_to = "THICKNESS")
```

The acquisition site is a three number code within the ID column prior to the _S_ identifier. Use this to create a new column containing only the acquisition site number to be used as a batch variable.

```
ADNI_metadata <- ADNI_metadata %>%
  separate_wider_position(ID,
    c(SITE = 3, 1, SAMPLE_ID = 6),
    cols_remove = FALSE)

ADNI_metadata <- as.data.frame(ADNI_metadata)
```

The scanner data is contained in the extra metadata file (not hosted on github, provided by Taki to Jessica). Using the ImageUID in the scanner data, combine this with the ImageID in the metadata and retain the scanner type (Manufacturer). Modify manufacturer to exclude spaces.

```
manufacturer_data <- ADNI_additional_metadata %>%
  select(imageUID, Manufacturer)
metadata_scanner <- merge(ADNI_metadata, manufacturer_data, by.x = "IMAGE_ID", by.y = "imageUID", all.x = TRUE)

for (i in 1:length(metadata_scanner$Manufacturer)){
  if(metadata_scanner$Manufacturer[i] == "GE MEDICAL SYSTEMS" ||
    metadata_scanner$Manufacturer[i] == "Philips Medical Systems"){
    metadata_scanner$Manufacturer[i] <- "GE_PHILLIPS"
  }
}

row.names(metadata_scanner) <- metadata_scanner$IMAGE_ID
```

Convert the columns to factors for downstream analyses.

```
metadata_scanner$SITE <- as.factor(metadata_scanner$SITE)
metadata_scanner$VISIT <- as.factor(metadata_scanner$VISIT)
metadata_scanner$DIAGNOSIS <- as.factor(metadata_scanner$DIAGNOSIS)
metadata_scanner$SEX <- as.factor(metadata_scanner$SEX)
metadata_scanner$Manufacturer <- as.factor(metadata_scanner$Manufacturer)
```

Create a summarized experiment object.

```
ADNI_se <- SummarizedExperiment(assays = list(cortical_thickness = specific_location_data),
  colData = metadata_scanner, metadata = list(metadata = metadata_scanner))
```

Results

Simulation Data

ADNI Data

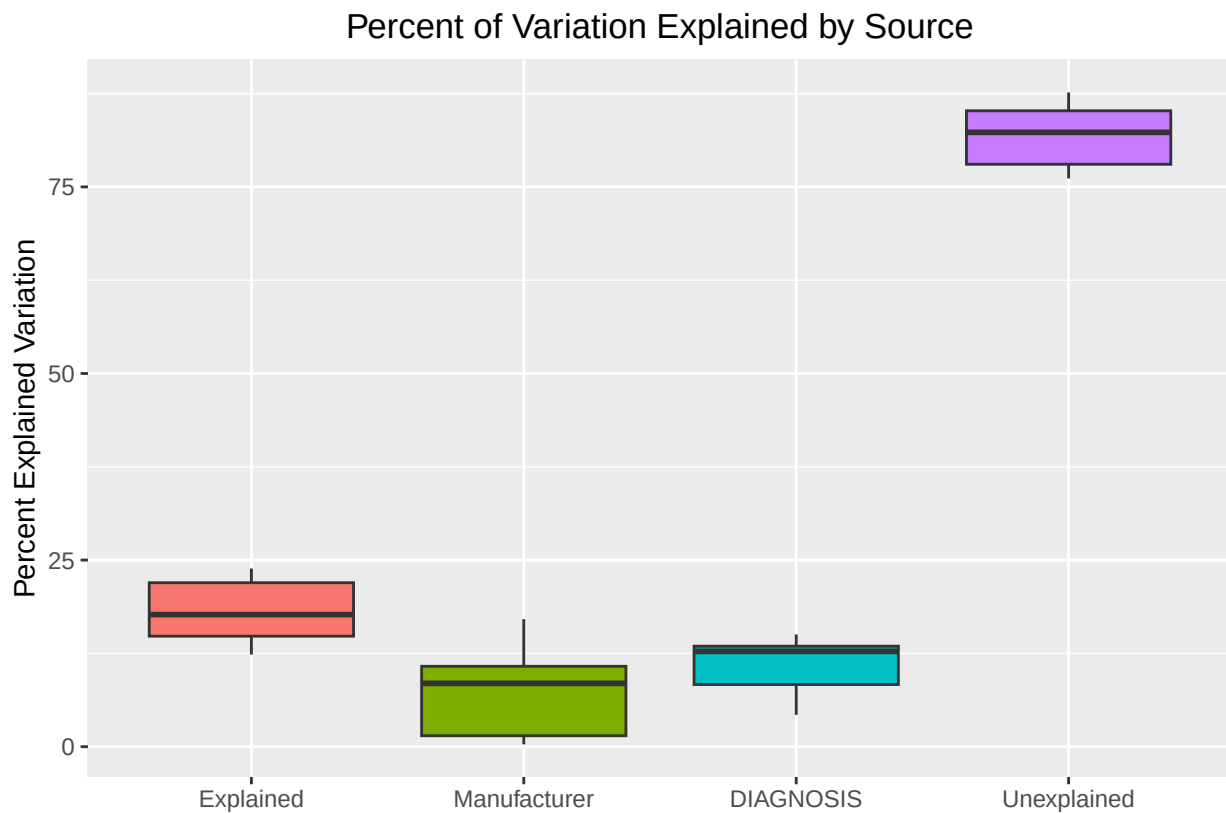
Evidence of Batch Effect Using BatchQC, we can view the raw data and identify evidence of batch effect within the data.

Below, we set up the variables that we are interested in analyzing: the raw data assay, the Manufacturer machine is the variable we suspect a batch effect to reside, and the diagnosis variable is where we would expect true differences between samples to be found.

```
cortical_assay <- "cortical_thickness"  
batch <- "Manufacturer"  
covar <- "DIAGNOSIS"
```

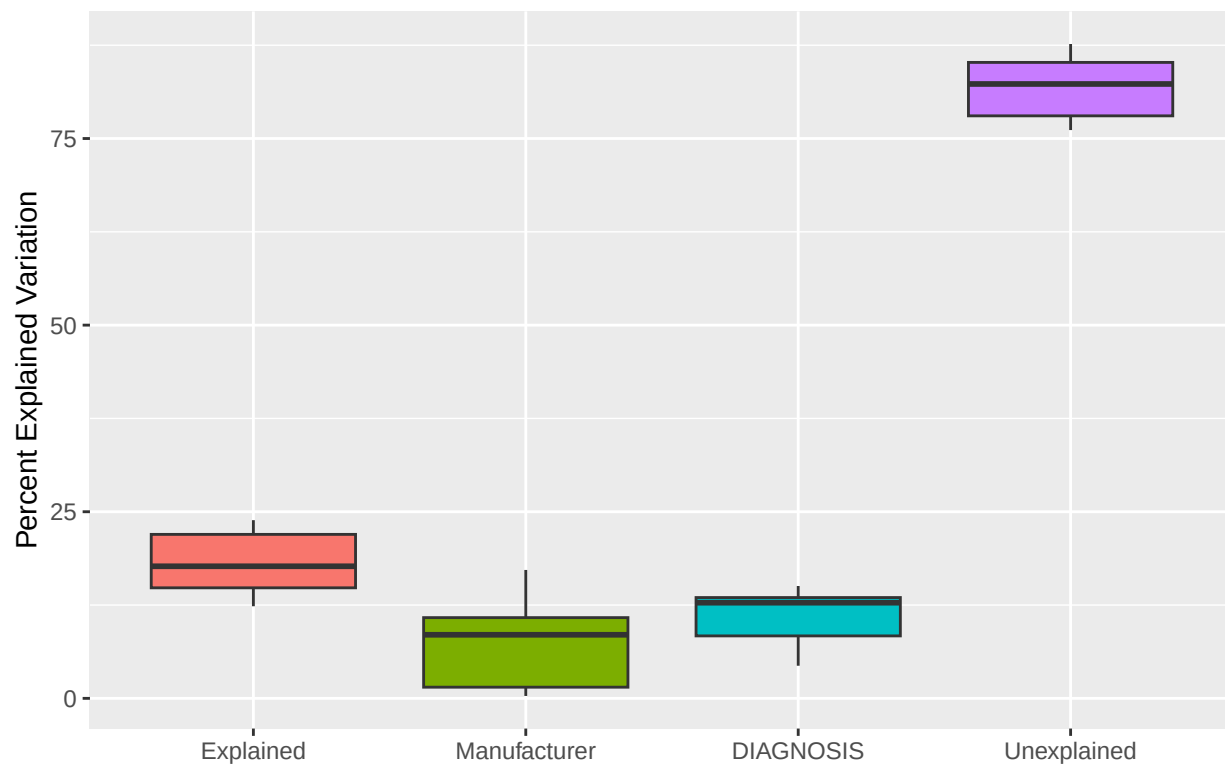
We plot a PCA plot of the raw data where color represents batch and shape represents the diagnosis. The ellipses shows the batch groupings and the distinct ellipses indicate that there is a batch effect within the data.

```
ex_variation <- BatchQC::batchqc_explained_variation(se = ADNI_se,  
  batch = batch,  
  condition = covar,  
  assay_name = cortical_assay)  
# individual variation  
EV_boxplot <- BatchQC::EV_plotter(batchqc_ev = ex_variation[[1]])  
EV_boxplot
```



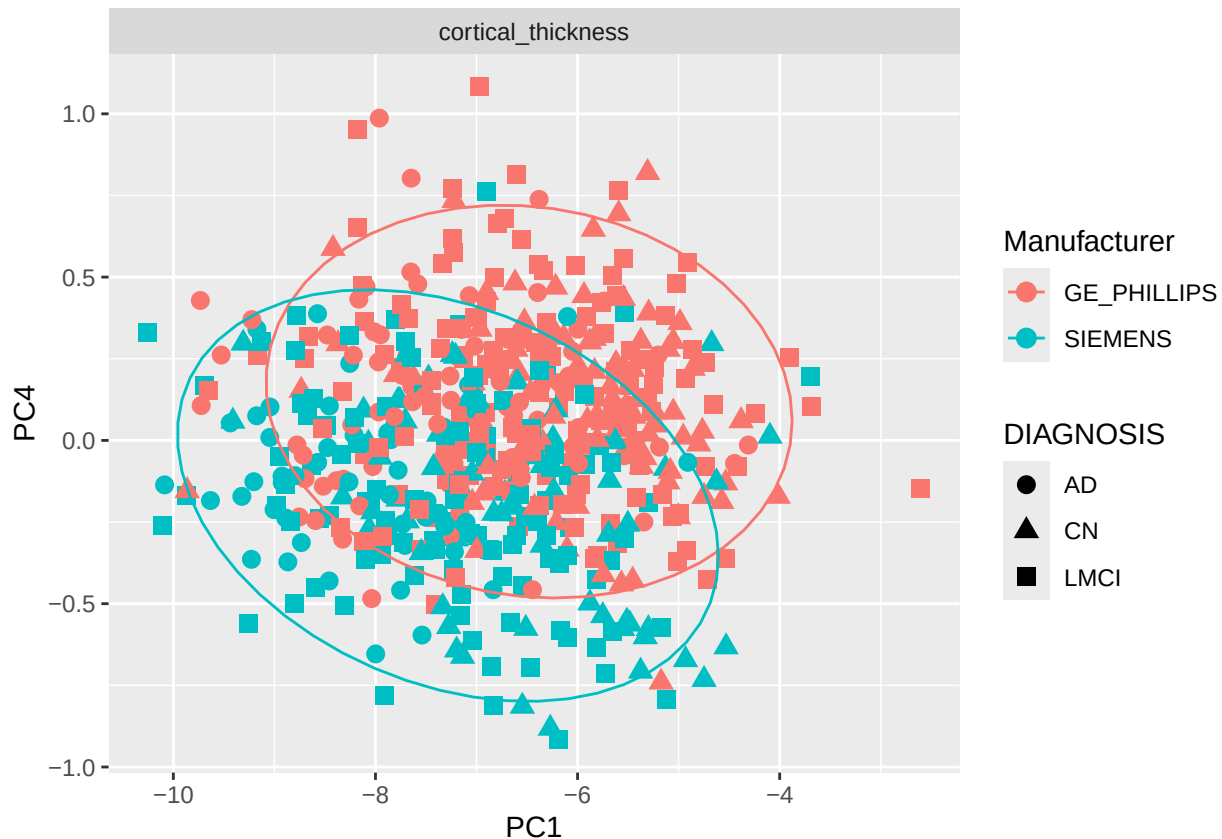
```
# Type 2 variation  
EV_boxplot_type_2 <- BatchQC::EV_plotter(batchqc_ev = ex_variation[[2]])  
EV_boxplot_type_2
```

Percent of Variation Explained by Source



```
pca_plot <- BatchQC::PCA_plotter(se = ADNI_se,
  nfeature = 8, color = batch,
  shape = covar, batch = batch,
  assays = cortical_assay,
  xaxisPC = 1, yaxisPC = 4, log_option = FALSE)
pca_plot$plot
```

```
## Warning: The following aesthetics were dropped during statistical transformation: shape.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
##   variable into a factor?
```



Lambda Statistic to Confirm need for Batch Correction We can confirm the need for a batch correction using our new tools. First, we will determine if our experiment is a balanced or unbalanced design.

Next, we calculate the lambda statistic

Based on this result...

Check Model Fit We will use our AIC model to check the fit of the data.

Since..., we can proceed with....

ComBat Batch Correction Use BatchQC to correct for batch effect using ComBat.

```
ADNI_se <- BatchQC::batch_correct(se = ADNI_se,
  method = "ComBat",
  assay_to_normalize = cortical_assay,
  batch = batch,
  covar = c(covar, "AGE", "SEX"),
  output_assay_name = "ComBat_cortical_thickness")
```

```
## Warning in formula.character(object, env = baseenv()): Using formula(x) is deprecated when x is a character vector.
## Consider formula(paste(x, collapse = " ")) instead.
```

```
## Found2batches
```

```
## Adjusting for1covariate(s) or covariate level(s)
```

```
## Standardizing Data across genes
```

```
## Fitting L/S model and finding priors
```

```
## Finding parametric adjustments  
## Adjusting the Data  
ComBat_assay <- "ComBat_cortical_thickness"
```

Other Data